E2 | Teaching the Competitive Firm

The animations introduce the ideas. Students attempt the exercise individually, compare briefly, and work through the answers with you on the whiteboard. The outside example opens a question that the final board problem resolves. No exercise prompts, worked answers, or new exercise scenes are added to the animation.

Use Exercise E2 throughout class. It uses the animation's firm with different prices: $TC = 1000 + q^2$, $MC = 2q$, $AVC = q$. All exercise work stays on the board. The original toffee-tax activity remains in the Part E practice bank.

Sequence for a 50 minute class

This is a working time budget, not a video runtime. The stop IDs below are verified against `E2_Animation.py`, scene `EpisodeE2`; the older storyboard and notes do not match every current hold.

Minutes	Animation or board work	Purpose
0–2	Opening story and board question	Establish the episode's question.
2–7	Play through `1.e.mr`	Market price, price taking, and why MR equals price.
7–9	Hide projection; Q1a	Retrieve MR with a different price.
9–16	Resume through `2.e`	Profit, the next-unit comparison, and the chosen quantity.
16–21	Hide projection; Q1b–c	Choose output and calculate profit.
21–28	Resume through `3.e`	Entry, losses, and operating versus shutting down.
28–32	Hide projection; Q2a	Compare two losses at a new price.
32–35	Resume through `3.f`	Long-run exit and zero economic profit.
35–37	Hide projection; Q2b–c	Distinguish exit from shutdown; retrieve entry.
37–47	Keep projection hidden; Q3	Apply the theory to the fixed-cost shock on the board.
47–49	Return to the opening question	Students explain the whole adjustment.
49–50	Resume to `4.a`	Preview downward-sloping firm demand and E3.

If time slips, shorten the numerical debrief after students have the correct calculation. Protect Q3 and the closing explanation. In a longer class, add the transfer question below rather than a second shock walkthrough.

Opening question

Before starting the animation, say:

> In 2006, an E. coli outbreak was linked to spinach. California introduced a leafy-greens food-safety program the following year, and growers surveyed afterward reported higher food-safety costs. That gives us a question about competition: how can a higher cost push some firms out of a market while the firms that stay eventually produce more?
>
> We'll isolate one kind of cost: a recurring expense that doesn't depend on how much a firm produces. We'll come back to that question once we have the pieces.

Write **Higher fixed cost: fewer firms, but more output per surviving firm?** in a corner of the board. Leave it there. Use "the firm" and "the market" through the rest of the lesson; no agricultural detail is needed in the exercise.

Source: UC Small Farms Network. The report documents higher seasonal food-safety costs. The pure recurring fixed expense and resulting exit are the classroom model, not claims about measured spinach-industry outcomes.

Running the room

At each stop, pause playback and hide the projection without advancing. Name the exercise part and put its givens on the board. Students have the prompt on paper.

Give everyone 30–60 seconds alone, then 30 seconds with a neighbor. Ask for a decision and a reason; build the board solution from their answers. Students correct their own work. For Q3, allow two minutes alone and one minute together before discussing. If answers diverge, offer one targeted cue and require another individual answer before explaining.

Spoken wording corrections

At `3.e.avc`, the screen says firms "exit" below AVC. Say **shut down in the short run**. The usual shutdown threshold is minimum AVC; at equality the firm is indifferent. At `3.e.fc`, replace "exiting and paying the fixed cost" aloud with **shutting down and still paying this period's fixed cost**. Long-run exit avoids future fixed costs. The animation is unchanged.

This firm's $AVC = q$ has a minimum of zero. The positive-price loss example demonstrates continued operation, not a positive shutdown-price threshold.

Board stops and spoken transitions

After 1.e.mr

Say: "Now close the picture. At a price of 90, what does one more sale add to revenue?"

Q1a: write $TR(20) = 1800$, $TR(21) = 1890$, so $MR = 90$. Ask why the earlier units still sell for 90. The explanation is price taking, not just the multiplication. Then resume: "Knowing what one more unit brings in lets us compare it with what that unit costs."

After 2.e

Q1b–c: write $P = 90$, $MC = 2q$, and $TC = 1000 + q^2$. Students choose $q = 45$. At $q = 40$, $MC = 80 < MR$; at $q = 50$, $MC = 100 > MR$. Increase from 40; decrease from 50. At 45, revenue is 4050, cost is 3025, and economic profit is 1025.

Draw a small firm graph with MC, ATC, and the price line, locating output first and profit second. Ask: "Why don't we choose the bottom of ATC?" Quantity maximizes the difference between revenue and cost; minimum ATC is not the general output rule. Use $MC = MR$ on rising MC, subject to the shutdown decision.

Resume: "What does a profit like this invite other firms to do?" Use the animation to introduce the market adjustment.

After 3.e

Q2a: write $P = 40$, $q = 20$, $TR = 800$, $VC = 400$, and $FC = 1000$. Students compare operating profit of -600 with shutdown profit of -1000. Operating contributes 400 toward an unavoidable expense. Ask for both losses before accepting "keep operating."

Resume: "That tells us what to do this period. It doesn't tell us whether to stay in this industry indefinitely."

After 3.f

Q2b–c: if the price of 40 persists, exit once future fixed costs can be avoided. If instead the market has the profitable price of 90 from Q1, firms enter, supply increases, and price falls. In the standard identical-firm model, adjustment ends at zero economic profit, $P = MC = \min ATC$.

Say: "We now have both decisions: how much to produce while operating, and whether to remain in the market. Let's use them together." Stay at the board for Q3. Do not return to an animated solution.

Final board problem

Q3 resets the market to long-run equilibrium, not the price of 40 or 90. The recurring fixed expense raises fixed cost from 1000 to 1600. This period's payment is unavoidable; future payments can be avoided by exiting. Demand and input prices stay fixed; firms are identical and entry and exit are free.

Students attempt both graphs before discussion. Build the answer in three stages:

1. **Before the shock.** Price meets MC at the minimum of ATC. Label P_0, q_0, and Q_0.
2. **Immediately after.** Raise ATC by $600/q$, a smaller gap at larger output, not a parallel shift. Leave MC and AVC fixed. Market supply, price, firm output, and total output initially stay fixed. Shade the loss between the old price and new ATC at q_0. Profit falls from zero to -600.
3. **After exit.** Shift market supply inward. Price rises; survivors move up unchanged MC and produce more. The market moves up demand and total output falls. Stop at the minimum of new ATC: zero economic profit, with fewer firms.

Ask someone to reconcile higher firm output with lower market output. Write $Q = Nq$: the decline in the number of firms more than offsets the increase in output per survivor. The market price rises through exit; individual price-taking firms do not choose to raise it.

Instructor check: initially $q_0 \approx 31.62$ and $P_0 \approx 63.25$; finally $q_1 = 40$ and $P_1 = 80$. Immediately after the expense, $TR = 2000$ and $TC = 2600$. Students need not calculate the minima.

Resolve the episode question

Point to the opening question. Give students 30 seconds alone to finish:

> A higher recurring fixed cost leaves the next unit's cost unchanged, so initially _____. It reduces _____, so over time _____. That raises _____, which leads the surviving firms to _____.

Expected: output is unchanged; profit falls; firms exit; market price rises; survivors produce more. Close with:

> A recurring cost can make staying in the industry less attractive without making the next unit more expensive. Exit changes the market price, and that changes the remaining firms' production decisions. This is one mechanism that can matter when businesses face new compliance costs.

Optional transfer: "Would a one-time payment already made, with no future payment required, create the same ongoing incentive to exit?" No: exiting cannot recover a sunk payment, and future operating costs have not changed.